EE4047 – High Voltage Techniques I

Study Questions:
Pre-midterm

Topic 1. Fundamentals of High Voltage Engineering

1. What are the main parts of interconnected power systems?
2. What is peak demand, how it changes annually in Turkey?
3. What is installed power, how is it compared to peak demand?
4. Where are the hydropower, thermal, wind and geothermal power plants mostly located geographically in Turkey?
5. What are the four sources with the highest share in electricity generation in Turkey?
6. Compare the overall length of transmission lines with distribution lines in Turkey.
7. How and why high voltage started to be commonly preferred for which reasons in the history of electricity?
8. How electric power that can be transmitted on an overhead AC line changes with voltage?
9. Why we do not prefer much higher voltages than the ones we are using today?
10. What are the origins of overvoltages?

11. Where and why HVDC is preferred?

12. Which voltages are high voltage? What are the subcategories of high voltage?

13. What are the types of HV components? Give three examples for each type.

Topic 1. Fundamentals of High Voltage Engineering

1. Please describe what is potential. What is its symbol? Draw and show on an example figure.

2. Please describe what is voltage. What is its symbol? Draw and show on an example figure.

3. What is alterating voltage? Plot an example waveform.

4. What is peak value? Show on an example figure.

5. What is rms value? Write its calculation formula and show on an example figure.

6. What is frequency?

7. What is period? Show on an example figure.

8. Why do we prefer three phase in which applications? What is neutral and ground? Draw an example figure.

9. What is an impulse voltage? Draw its waveform. Show the peak, front and tail of the impulse.

10. How are the timings for lightning and switching impulse voltages? How are they calculated? Show on an example figure.

11. Describe transformers and why there are used in HV systems.

12. Describe generator types and how they are used in HV systems.

13. Describe circuit breaker types and how they are used in HV systems. Compare different circuit breaker types with each other from several aspects.

14. Describe disconnector types and how they are used in HV systems.

15. Write the names of pylon types. Draw a pylon and show its main parts.

16. Describe isolator types and how they are used in HV systems.

17. Draw a cable and show its parts. Write the different materials used in cables.

18. Why buses used in HV systems?

Topic 1. Fundamentals of High Voltage Engineering

19. What are current and voltage transformers? Why are they used in HV systems?
20. What are main types of voltage dividers? Why are they used in HV systems?
21. What is surge arrester? Why is it used in HV systems?
22. What is arcing horn? Why is it used in HV systems?
23. What is corona ring? Why is it used in HV systems?
24. What is GIS substation? Why is it used in HV systems?

Topic 2. Generation of High Voltages

1. What are the two main methods to generate HVDC?

2. What are the two main methods to generate AC high voltage?

3. How high voltage can be generated based on AC to DC conversion?

4. How HVDC is generated using single phase half-wave rectifier? Draw an example circuit and explain it.

5. How HVDC is generated using single phase full-wave rectifier? Draw an example circuit and explain it.

6. Why cascaded rectifier circuits are used? Draw an example and explain it.

7. What is electrostatic generator? How it Works? Draw an example diagram and explain it.

8. Draw and explain the parts of two main types of high voltage testing transformers.

9. Why cascaded transformers are used? Draw an example cascaded transformer and explain power and voltages change at each stage.

Topic 2. Generation of High Voltages

10. What is series resonant circuit? How it generates AC high voltages? Draw an example and explain it.

11. What is impulse voltage? Show an example impulse voltage waveform and show the main parts on the figure. What are the two types of impulse voltage? What are timings? How the timings are calculated?

12. What is single-stage impulse voltage generator circuit? What are its parts and how it works? Draw an example circuit and explain it.

12. What is multistage impulse voltage generator circuit? Why is it used and how it works? Draw an example figure and explain.

Topic 3. Electrostatic Fields and Planar Electrode Systems

1. What is electrostatic field? What is its difference from magnetostatic field and electromagnetic field?

2. What are cartesian, cylindrical and spherical coordinate systems? Explain example figures for each and explain.

3. What is scalar multiplication of two vectors? Show an example.

4. What is vectoral multiplication of two vectors? Show an example.

5. What are equipotential lines? How they are found and drawn?

6. What are electric field lines? How they are drawn?

7. What is a planar electrode system? What does it represent in reality? Draw an example panar electrode system.

8. How voltage and electric field changes through the gap between the electrodes in a planar electrode system?

Topic 3. Electrostatic Fields and Planar Electrode Systems

9. How a discharge or breakdown can be understood in a system using electric field strength?

10. Solve Example-1 in Week 5.2 Tuesday 22.03.22 lesson slides using different numerical values.

11. Draw equipotential lines and electric field lines in an example planar electrode system.

12. Draw an example multilayer dielectric planar electrode system, showing how the representative capacitances are connected to each other.

13. Draw equipotential lines and electric field lines in an example concentric spherical electrode system.

Topic 4. Spherical Electrode Systems

1. What is a concentric spherical electrode system? What does it represent in reality? Draw an example concentric spherical electrode system.

2. How voltage and electric field changes through the gap between the electrodes in a concentric spherical electrode system?

3. How voltage and electric field changes through the gap between the electrodes in a concentric spherical electrode system?

4. What are actual electrode seperation, effective electrode seperation, geometric characteristic, utilization factor (electrode efficiency)?

5. How breakdown differs under constant r_2 and U, due to changing r_1, based on $^{r_1}/_{r_2}$? Show on an example graph.

6. Solve all the examples provided in Week 7.1 Monday 04.04.22 slides using different numerical values.

7. What are possible spherical electrode configurations other than the concentric configuration?

How electrode gap can be represented in such systems in a similar manner with concentric configuration?